

# MEVIN JOSE

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## Summary

AI/ML undergraduate specializing in agentic systems, evaluation-driven ML pipelines, and explainable RL. Built a multi-agent SQL system with 95-100% accuracy on a 100-query benchmark with schema-grounded validation, a PPO traffic optimization system improving emergency transit time by 10.6%, and real-time CV pipelines sustaining <100ms latency. Focused on system design, validation rigor, and production-ready ML workflows.

## Skills

**Languages:** Python, SQL, JavaScript, TypeScript, Java

**ML & AI:** PyTorch, Scikit-learn, LightGBM, OpenCV, SHAP, NLP

**LLM & Agentic Systems:** LangChain, LangGraph, RAG, FAISS, Sentence Transformers, Prompt Engineering, Multi-Agent Systems, Hugging Face, PEFT (LoRA), RAGAS

**MLOps & Infrastructure:** FastAPI, Docker, GitHub Actions, AWS (EC2, S3), Drift Detection, Model Monitoring, CI/CD

**Databases & Tools:** PostgreSQL, MongoDB, Redis, SQLite, Pinecone, Qdrant, Flask, Streamlit, Pandas, NumPy, SciPy

## Work Experience

### Wadhvani Foundation

Bangalore, India

AI Intern

Jun 2026 – Present

- Participated in technical architecture reviews for an enterprise AI document ingestion platform, analyzing RAG retrieval abstractions and vector search flows to optimize multi-tenant query routing.
- Evaluated open-source multi-modal embedding models to replace legacy Qwen models, benchmarking Qdrant vector retrieval performance and cost efficiency.

### Superhero Learning

Bangalore, India

Web Developer Intern

Jun – Sep 2025

- Deployed 8+ MERN pages for 100+ active users, optimizing MongoDB indexing and query plans to reduce API latency by 25%.
- Improved load speed by 10% and cut iteration time by 25% by architecting performance observability pipelines and asset compression workflows.

## Projects

### AI Data Analyst Agent

Python, FastAPI, FAISS, Sentence Transformers, Groq/Gemini APIs, RAG, Pydantic

- Achieved 95-100% natural-language-to-SQL accuracy on a 100-query benchmark spanning 8 e-commerce domains, validated against schema-grounded ground truth.
- Architected planner-executor-evaluator multi-agent system with FAISS schema retrieval and multi-model fallback (Groq → Gemini), improving execution accuracy.
- Integrated Dockerized execution sandboxes to isolate generated SQL scripts, reducing unsafe execution risks to 0% during schema validation.

### SentinelGraph – Explainable Anomaly Detection Engine

Python, Streamlit, NetworkX, RAGAS, PyTorch, Docker

- Architected a 7-detector behavioral anomaly ensemble for 1,000+ entities, suppressing false positive alert rates by >90% via max-dominant risk fusion.
- Evaluated GNN and Graph-RAG traversal pipelines with RAGAS, achieving 0.942 PR-AUC and sub-10ms scoring latency over 10K benchmark events.
- Integrated counterfactual explainability and SHAP feature breakdowns, increasing policy explainability and threat triage efficiency by 30%.

### Adaptive Traffic Signal Control (PPO)

PPO, SUMO, Stable-Baselines3, SHAP

- Reduced emergency vehicle transit time by 10.6% vs fixed baseline using PPO-based reinforcement learning.
- Applied SHAP-based interpretability over 43D state space, increasing policy reliability and explainability by 30% in safety-critical simulation scenarios.

## Education

### PES University

Bangalore, India

B.Tech, Computer Science (AI & Machine Learning)

Aug 2023 – May 2027

CGPA: 7.85

**Coursework:** Explainable AI, LLMs & Applications (RAG), Advanced Foundations of ML (Deep Learning, Reinforcement Learning), Machine Learning, Optimizations & Metaheuristics, Design and Analysis of Algorithms, Python

## Certifications

Google Data Analytics — Meta GenAI — Microsoft Computer Vision — Kaggle ML & DL